

OPTIMIZING RETAIL PRICING STRATEGY USING BEHAVIORAL PRICE SIGNALS

A data-driven pricing advisory framework using online retail data

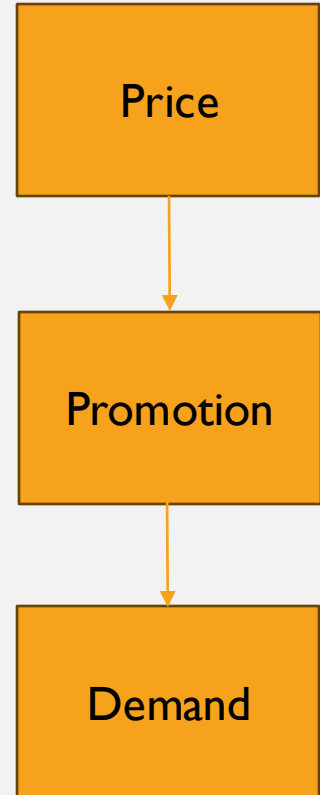
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MSBA | Retail Analytics | Pricing Strategy

Tools: Python, APIs, ML, Prescriptive Analytics

RETAIL PRICING DECISIONS ARE BEING MADE WITHOUT CLEAR DEMAND SIGNALS

- Retail prices change frequently, but **sales data is often unavailable**
- Promotions are applied broadly without knowing **which SKUs respond**
- Discounting risks **margin erosion** without guaranteed demand lift
- Managers need **decision support**, not just descriptive analytics



KEY BUSINESS QUESTIONS THIS ANALYSIS ADDRESSES

 How do retailers price in practice — stable or promotional?

 Can pricing behavior alone reveal **pricing strategy**?

 Which pricing actions correlate with **strong vs weak demand**?


 How can we test pricing decisions **before executing promotions**?

DAILY COMPETITIVE PRICING DATA ACROSS WALMART AND EBAY

UPC	Retailer	Product Title	Price (USD)	Capture Date
190199801561	ebay	Beats - Flex Wireless Earphor	49.99	2025-11-25
190199801561	walmart		30.01	2025-11-25
194252484296	ebay	Beats Fit Pro - True Wireless I	175.95	2025-11-25
194252484296	walmart		149.0	2025-11-25
194644039165	ebay	Soundcore K20i Semi-in-Ear E	29.99	2025-11-25
194644039165	walmart		19.99	2025-11-25
194644126155	ebay	Soundcore - by Anker P25i Tr	24.99	2025-11-25
194644126155	walmart		19.99	2025-11-25
194644186333	ebay	Soundcore P30i Noise Cancel	29.99	2025-11-25
194644186333	walmart		24.99	2025-11-25

- Daily price snapshots from **public retailer APIs**
- 173 matched SKUs (Headphones & Earbuds)
- Walmart (structured retailer) vs eBay (marketplace)
- Captures prices, discounts, promotions over time

RETAILERS EXHIBIT DISTINCT PRICING BEHAVIORS THAT CAN BE MEASURED

Behavioral Trait	Walmart (Structured Retailer)	eBay (Marketplace)
Price Stability	Varying	High
Volatility	Low-Medium	Low
Promo Activity	Moderate & structured	None
EDLP Consistency	Varying	High
Price vs MSRP	Anchored	Not Captured
Rigidity	Mixed	Strong

- Pricing strategies leave **behavioral fingerprints**
- Stability, volatility, and promotions reveal intent
- These patterns can be quantified without assumptions

PRICING STRATEGY CAN BE DIAGNOSED USING A SMALL SET OF BEHAVIORAL SIGNALS

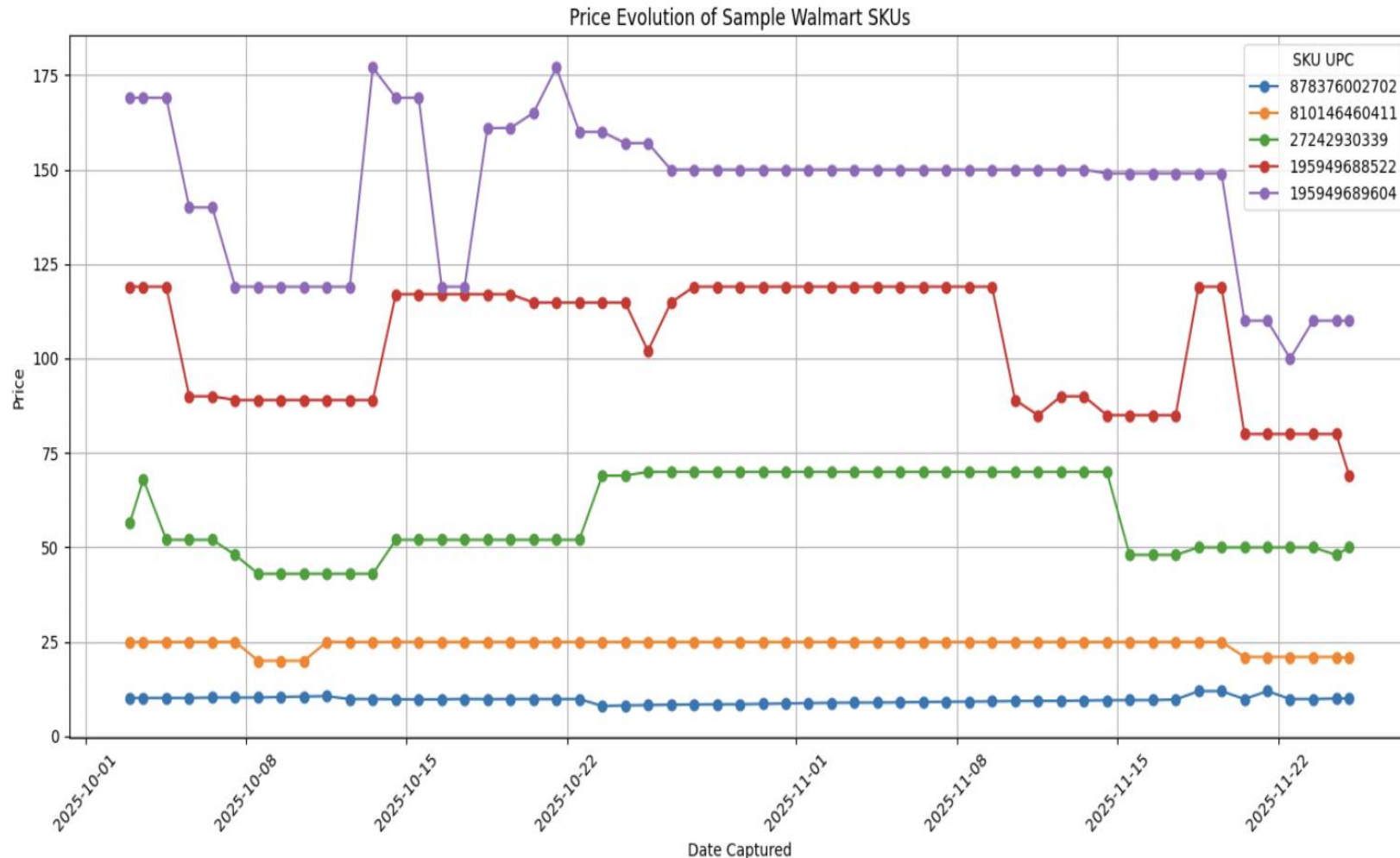
 **Stability signals:** rigidity,
EDLP consistency

 **Dynamic signals:** volatility,
price change frequency

 **Promotional signals:** promo
intensity, discount depth

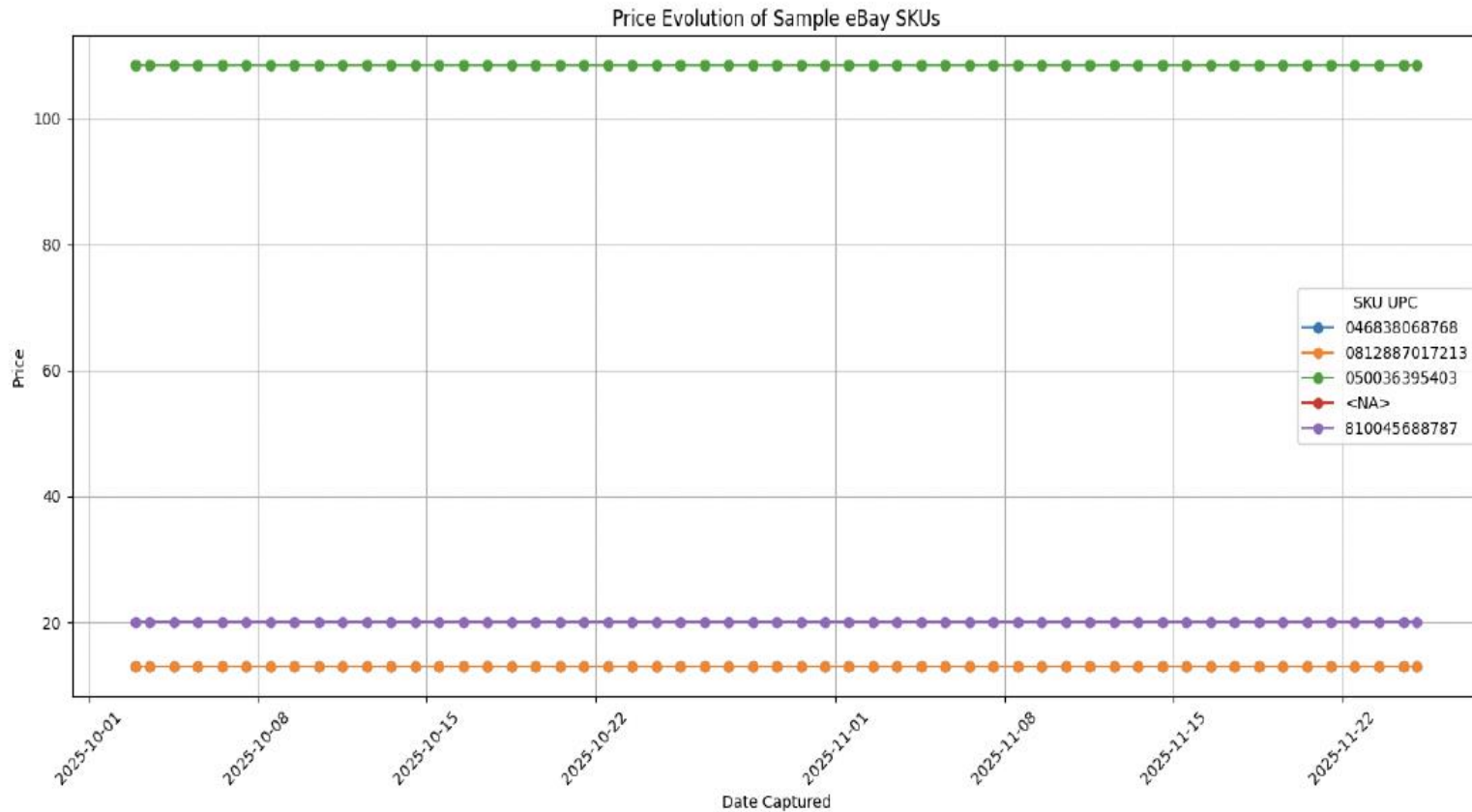
Together, these signals distinguish EDLP, Hybrid, and Hi-Lo behavior.

WALMART USES STABLE PRICING WITH SELECTIVE, STRUCTURED PROMOTIONS



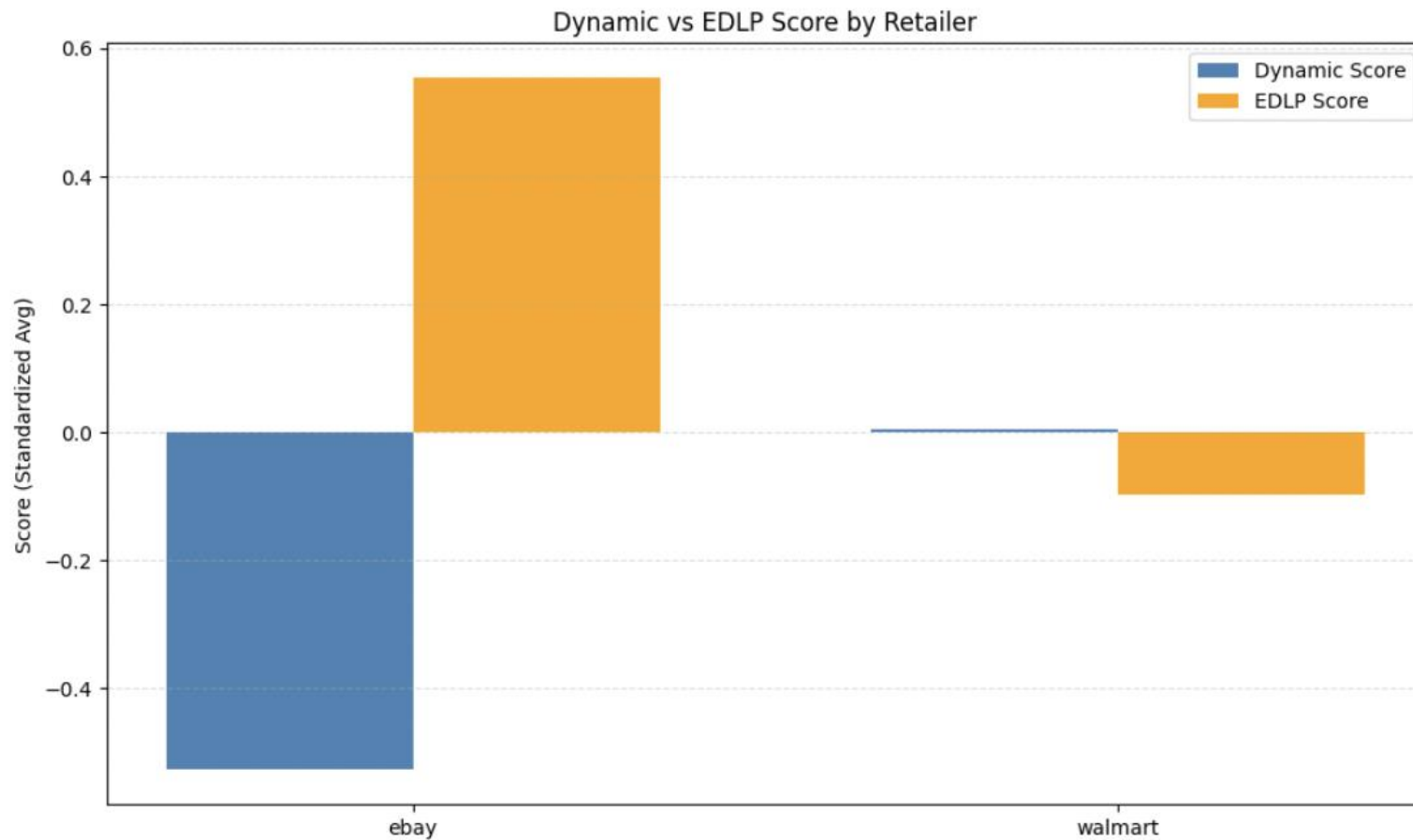
- Prices remain stable for most SKUs
- Promotions are intentional and SKU-specific
- Indicates a Hybrid pricing strategy

EBAY MAINTAINS HIGHLY STABLE PRICING WITH MINIMAL PROMOTIONAL ACTIVITY



- Prices rarely change
- Little to no discounting
- Reflects EDLP-like behavior driven by seller inertia

PRICING BEHAVIOR CLASSIFIES WALMART AS HYBRID AND EBAY AS EDLP

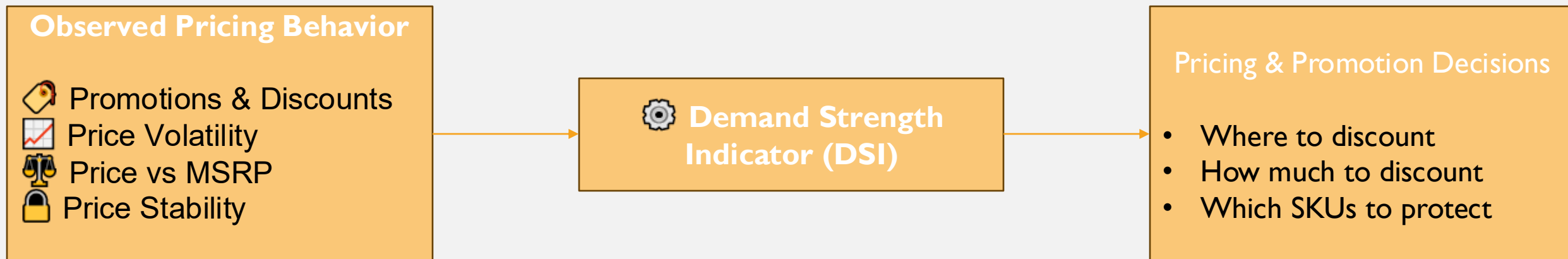


- Classification is **data-driven**, not assumed
- Based on observed price behavior
- Enables benchmarking and competitive positioning

DEMAND CAN BE INFERRED EVEN WITHOUT SALES DATA

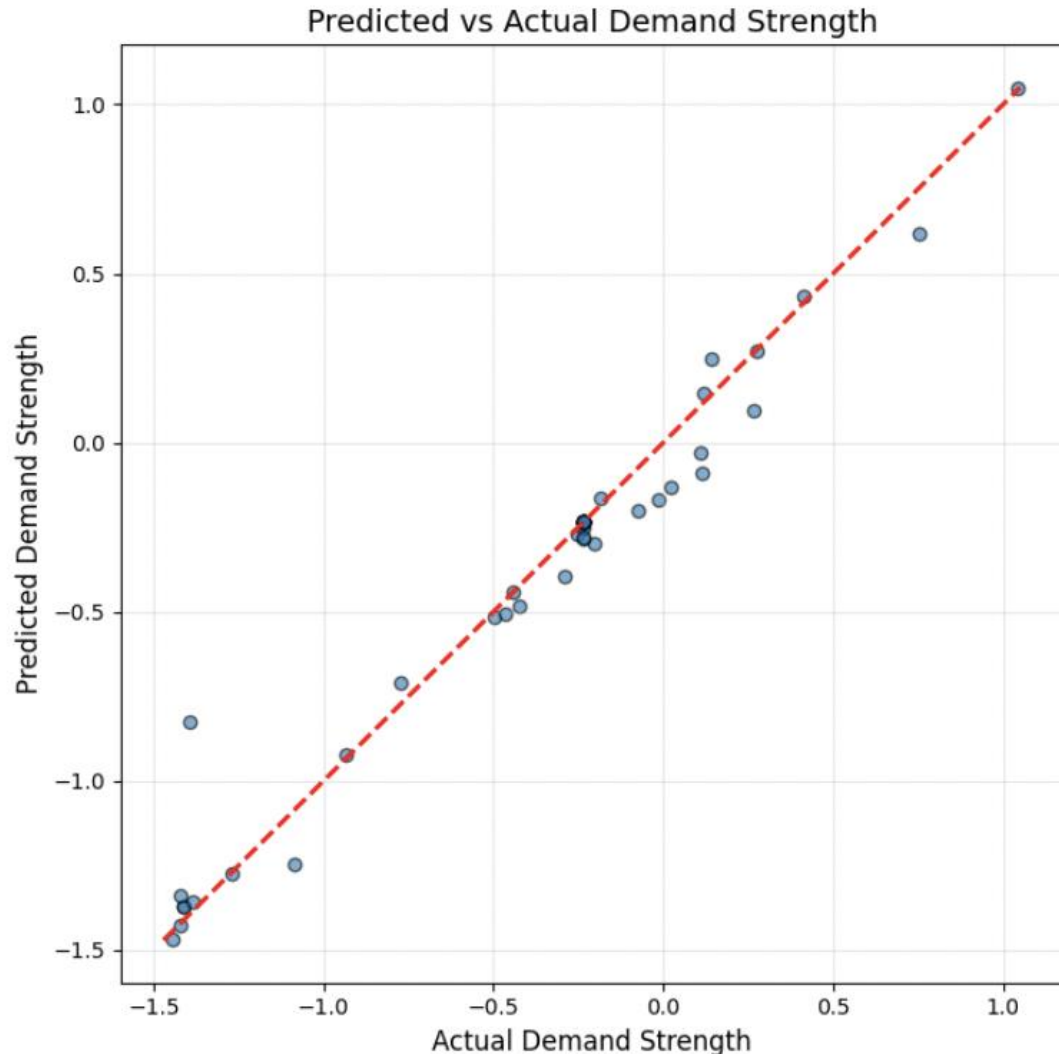
- Sales data often unavailable in competitive retail
- Pricing actions reflect **underlying demand**
- Built a composite **Demand Strength Indicator (DSI)**

DSI is: A relative measure of how “healthy” demand appears for each SKU.



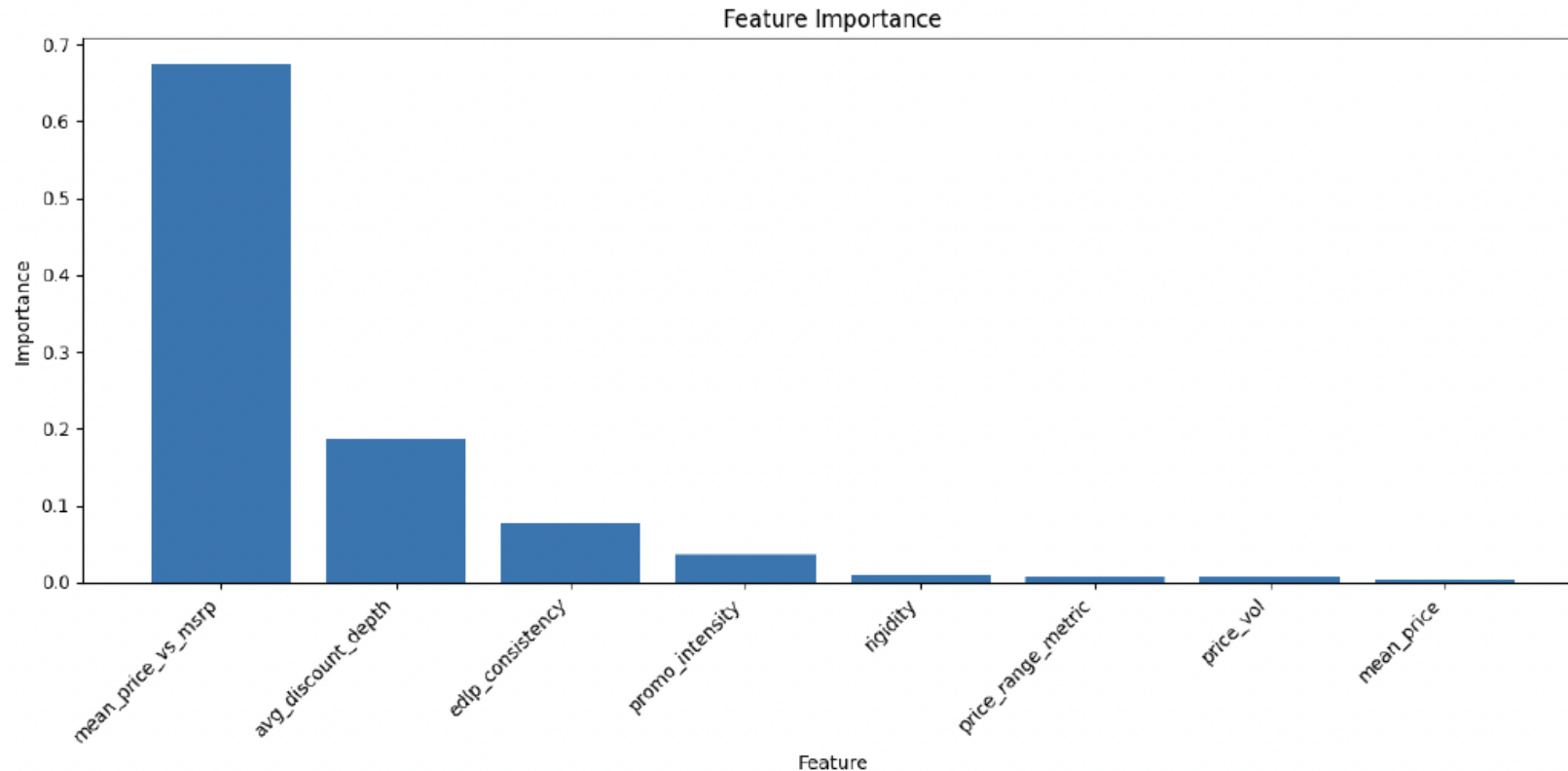
PRICING AND PROMOTION SIGNALS

PREDICT DEMAND STRENGTH WITH HIGH ACCURACY



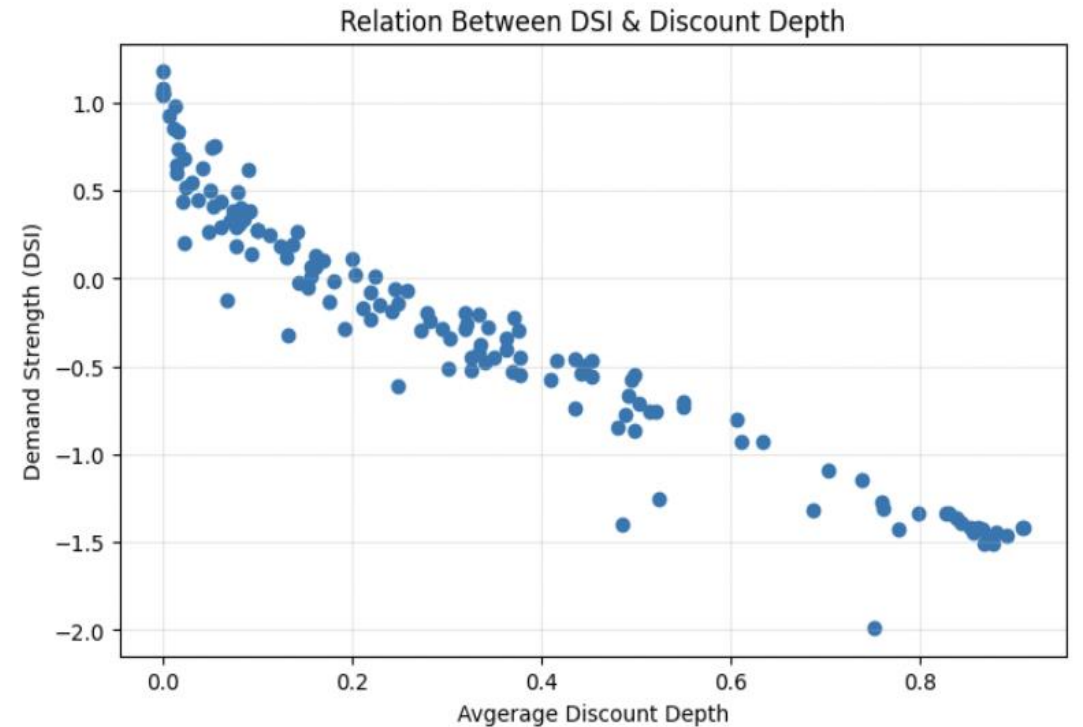
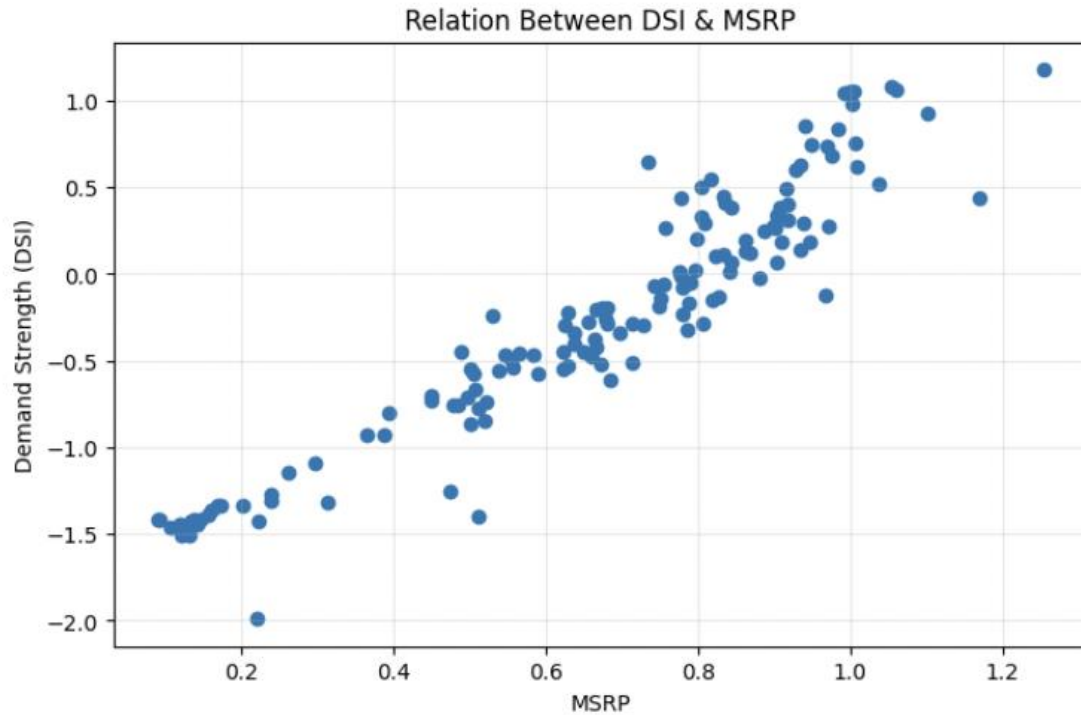
- ML model learns nonlinear pricing–demand patterns
- Strong predictive performance ($R^2 \approx 0.96$)
- Identifies **key demand drivers**

COMPETITIVE PRICE POSITIONING MATTERS MORE THAN DEEP DISCOUNTS



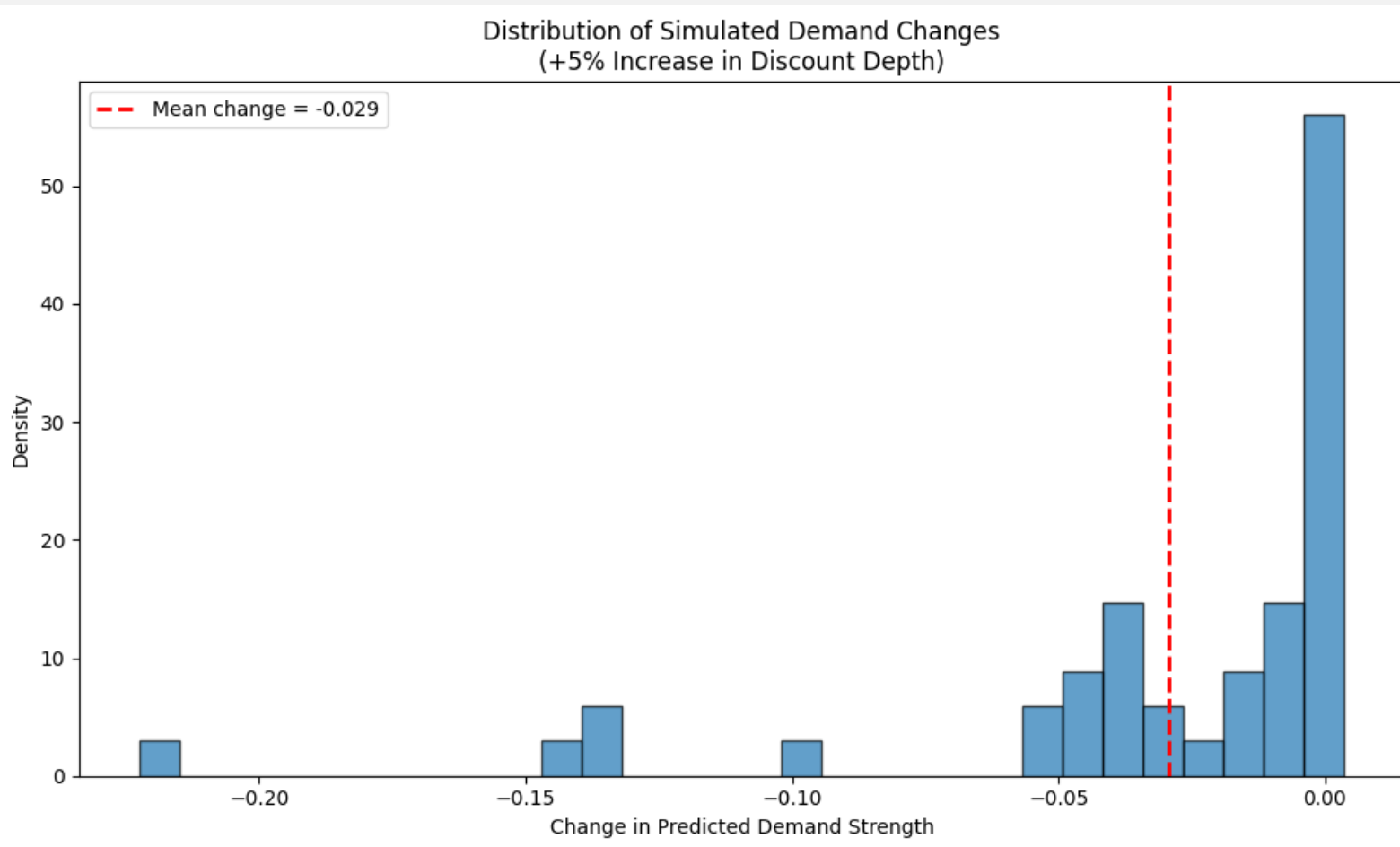
- Price vs MSRP is strongest driver
- Deep discounts often reflect weak demand
- Stability correlates with stronger demand signals

STABLE PRICING NEAR MSRP IS ASSOCIATED WITH STRONGER DEMAND SIGNALS



- SKUs priced close to MSRP show higher DSI
- Heavy discounting correlates with weaker demand

NOT ALL DISCOUNTS INCREASE DEMAND — SKU- LEVEL SENSITIVITY VARIES



- Simulated +5% discount impact
- Some SKUs improve, others worsen
- Blanket promotions are risky
- The average effect of increased discount depth is decrease in demand

DATA-DRIVEN PRICING ENABLES SMARTER PROMOTIONS AND MARGIN PROTECTION



Segment SKUs by discount
sensitivity



Protect margins on high-demand
SKUs



Avoid uniform promotions



Use simulations before
committing budget

PRICING DECISIONS CAN BE SIMULATED BEFORE CAPITAL IS COMMITTED



Better Pricing Decisions

Data-backed pricing actions



Reduced Margin Risk

Avoid value erosion from blanket discounts



Smarter Promotion Planning

Test scenarios before committing budget